

Trigonometric ratios from points



1

a $r = 17$ units

b

i	$\frac{8}{17}$	ii	$\frac{15}{17}$	iii	$\frac{8}{15}$	iv	$\frac{17}{8}$
v	$\frac{17}{15}$	vi	$\frac{15}{8}$				

2

a $r = 25$ units

b

i	$-\frac{24}{25}$	ii	$-\frac{7}{25}$	iii	$\frac{24}{7}$	iv	$-\frac{25}{24}$
v	$-\frac{25}{7}$	vi	$\frac{7}{24}$				

3

a $r = 25$ units

b

i	$\frac{24}{25}$	ii	$-\frac{7}{25}$	iii	$-\frac{24}{7}$	iv	$\frac{25}{24}$
v	$-\frac{25}{7}$	vi	$-\frac{7}{24}$				

4

a

i	$\frac{4}{5}$	ii	$\frac{3}{5}$	iii	$\frac{4}{3}$
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b

i	$\frac{20}{29}$	ii	$-\frac{21}{29}$	iii	$-\frac{20}{21}$
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c

i	$-\frac{21}{29}$	ii	$\frac{20}{29}$	iii	$-\frac{21}{20}$
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Identify quadrants

5 $\frac{-\sqrt{15}}{8}$

6



- a Quadrant 2
 b Quadrant 3
 c Quadrant 2
 d Quadrant 1
 e Quadrant 3 and 4
 f Quadrant 1 and 4
 g Quadrant 1 and 3
 h None
 i Quadrant 1
 j Quadrant 4
 k Quadrant 1
 l Quadrant 3

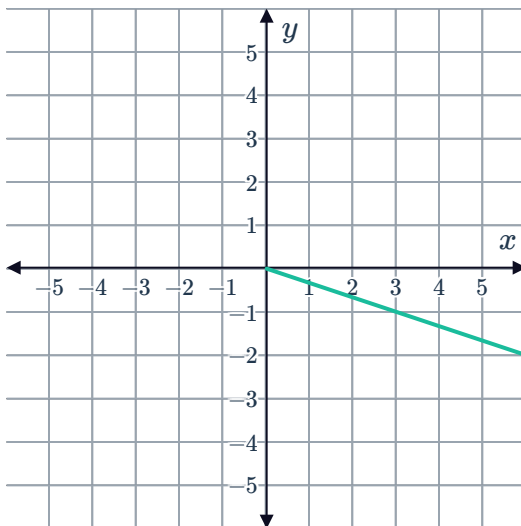
7

- a Quadrant 1 b Quadrant 4 c Quadrant 3 d Quadrant 2

Equation of the terminal side of an angle

8

a

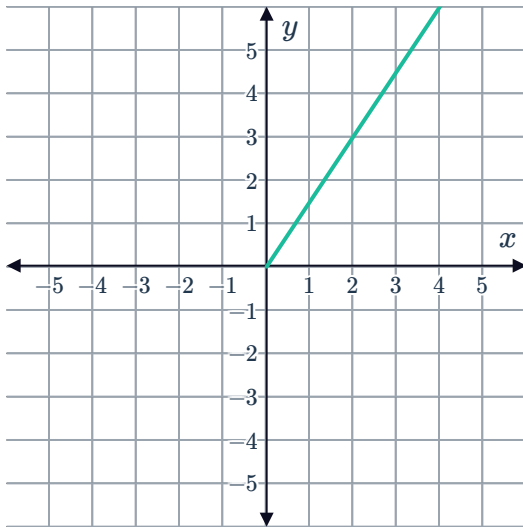
b $3, 1, \sqrt{10}$

c

- i $-\frac{\sqrt{10}}{10}$ ii $\frac{3\sqrt{10}}{10}$ iii $-\frac{1}{3}$ iv $-\sqrt{10}$
 v $\frac{\sqrt{10}}{3}$ vi -3

9

a

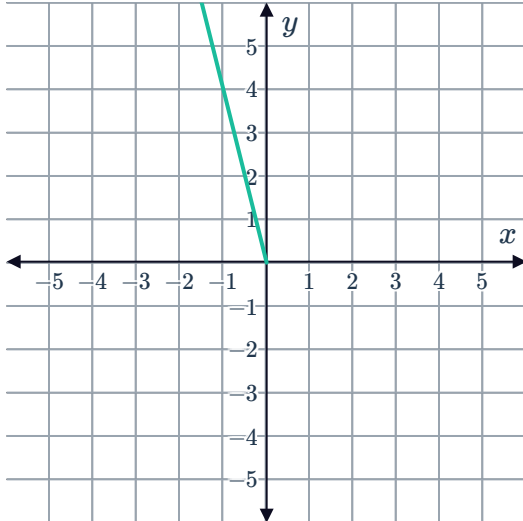
b 2, 3, $\sqrt{13}$

c

i $\frac{3\sqrt{13}}{13}$ ii $\frac{2\sqrt{13}}{13}$ iii $\frac{3}{2}$ iv $\frac{\sqrt{13}}{3}$
 v $\frac{\sqrt{13}}{2}$ vi $\frac{2}{3}$

10

a

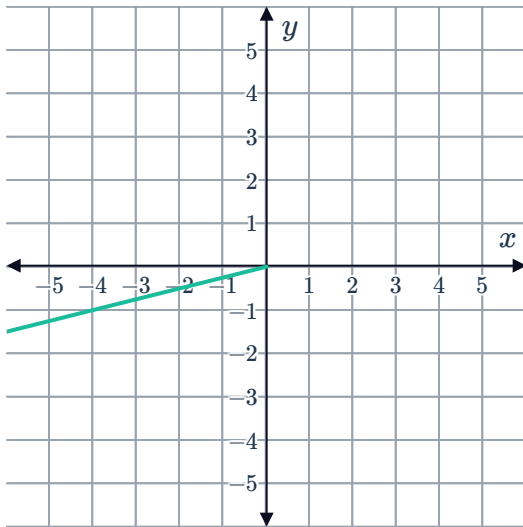
b 4, 1, $\sqrt{17}$

c

i $\frac{4\sqrt{17}}{17}$ ii $-\frac{\sqrt{17}}{17}$ iii -4 iv $\frac{\sqrt{17}}{4}$
 v $-\sqrt{17}$ vi $-\frac{1}{4}$

11

a

b 4, 1, $\sqrt{17}$

c

i $-\frac{\sqrt{17}}{17}$

ii $-\frac{4\sqrt{17}}{17}$

iii $\frac{1}{4}$

iv $-\sqrt{17}$

v $-\frac{\sqrt{17}}{4}$

vi 4